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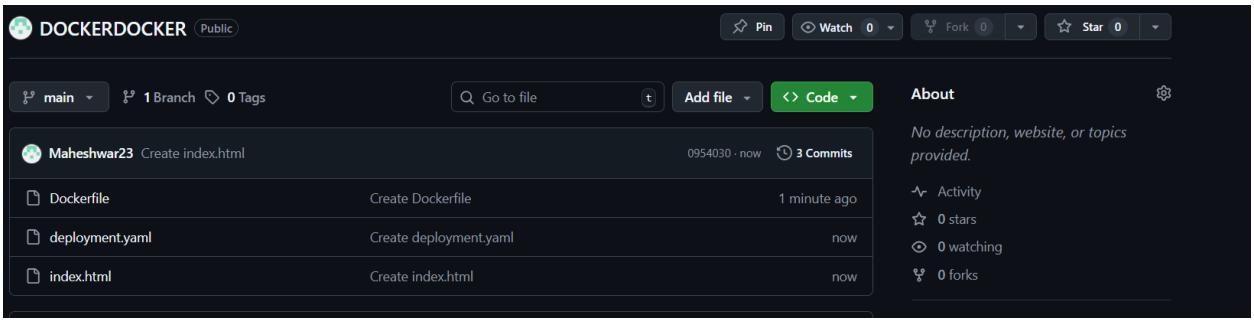
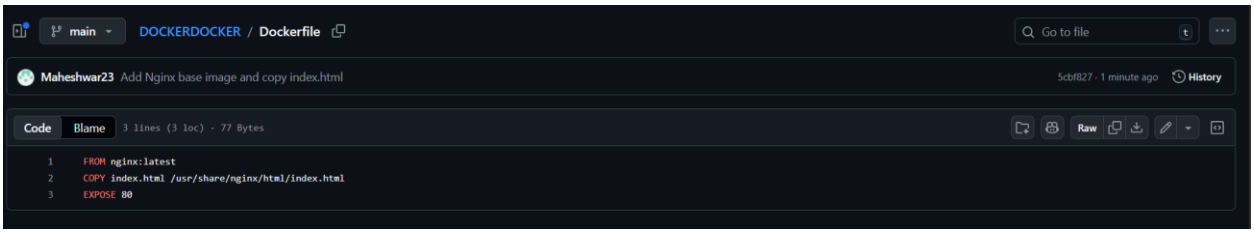
Agile and Devops Lab Assignment 4 Document

1. **Create a repository in GIT and create the following files:**
   - **index.html**
   - **Dockerfile**
   - **Deployment.yaml**





main DOCKERDOCKER / index.html

Go to file

Maheshwar23 Create index.html for CI/CD demo webpage b0e5755 · now History

Code Blame 21 lines (21 loc) · 363 Bytes Raw

```html
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center;
font-family:Times New Roman;
background-color:#af0d0d;
}
h1{
color:#0ef777;
}
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```

main DOCKERDOCKER / deployment.yaml

Go to file

Maheshwar23 Add Deployment and Service for html-demo application 12afc18 · now History

Code Blame 32 lines (32 loc) · 443 Bytes Raw

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: html-demo
spec:
 replicas: 2
 selector:
 matchLabels:
 app: html-demo
 template:
 metadata:
 labels:
 app: html-demo
 spec:
 containers:
 - name: html-demo
 image: swetab84/html-demo:latest
 ports:
 - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
 name: html-demo-service
spec:
 type: NodePort
 selector:
 app: html-demo
```

2. **Enable Kubernetes**

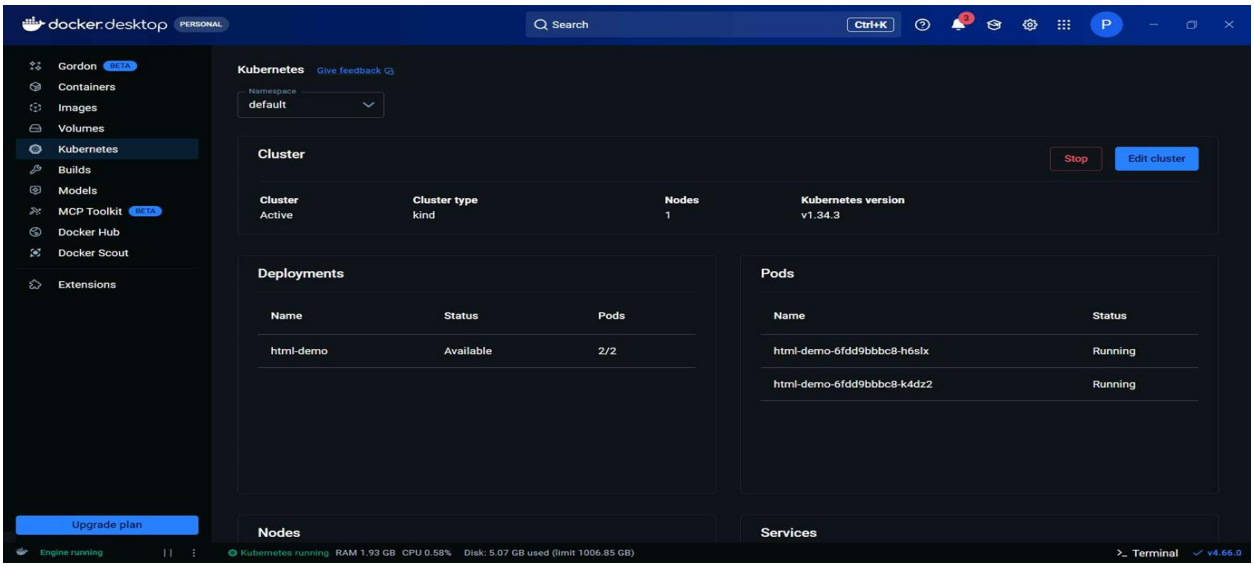
   • **Open the Docker Dashboard:** Launch the Docker Desktop application on your system.

   • **Navigate to Settings**: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.

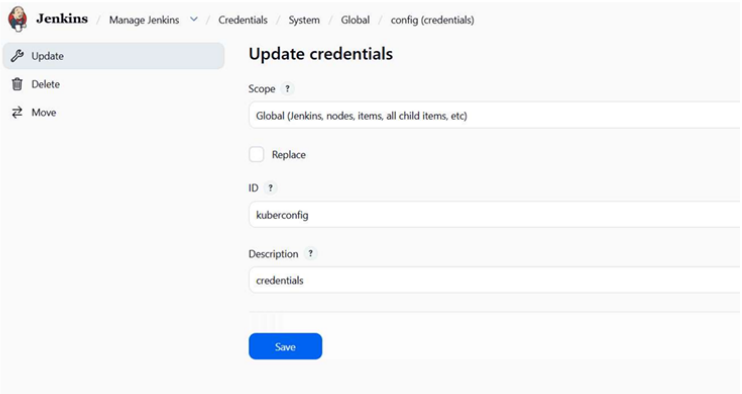
   • **Select the Kubernetes Tab**: In the Settings menu, find and select the **Kubernetes** tab.

   • **Enable Kubernetes**: Check the box labeled **Enable Kubernetes.**

   • **Apply & Restart:** Click the **Apply & Restart** button at the bottom of the settings window. Docker Desktop will then download the necessary **Kubernetes** components and start a single-node cluster. This may take a few minutes the first time.



New Credentials – Secret – Broswse C: User->Administrator-->.kube



Create New Item -> Pipeline Paste the code in pipeline script
- Update URL of GIT repository
- Make sure credentialsId: "dockerhub" same as mentioned in credentials
- Make sure that the ID mentioned in credentials "kuberconfig" is same as in script
- Run the Build

Jenkins / Docker Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline {
2 agent any
3 environment {
4 DOCKER_IMAGE = "maheshwar/html-demo"
5 }
6 stages {
7  stage('Clone code') {
8 steps {
9 git branch: 'main',
10 url: 'https://github.com/maheshwar/AglleLabAssessment5.git'
11 }
12 }
13 stage('Build Docker Image') {
14 steps {
15 bat 'docker build -t %DOCKER_IMAGE% .'
```

☑ Use Groovy Sandbox ?

Pipeline Syntax

Save Apply

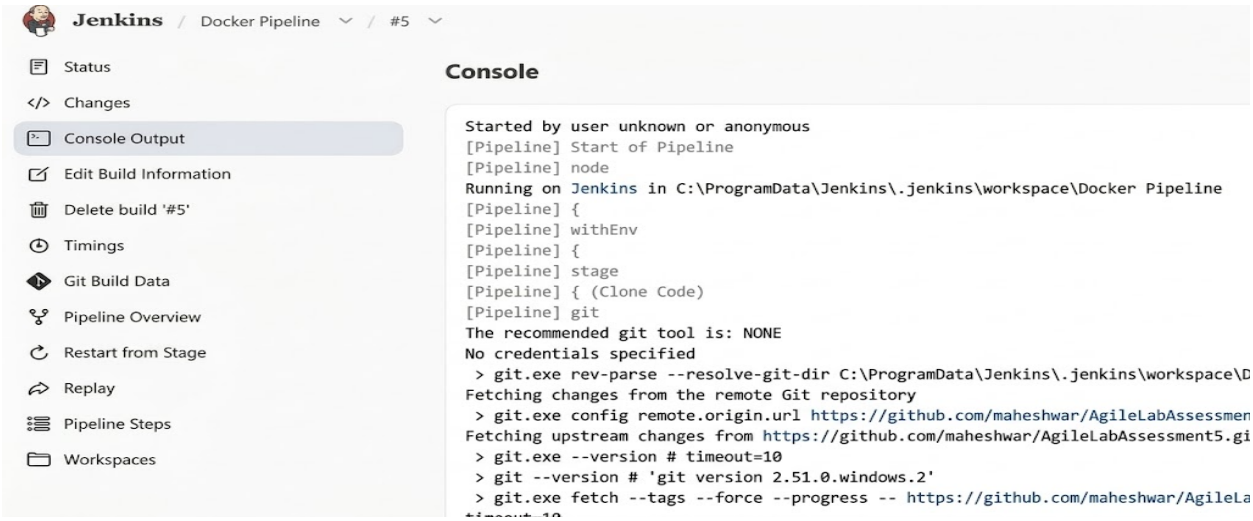
Docker Pipeline

Stage View

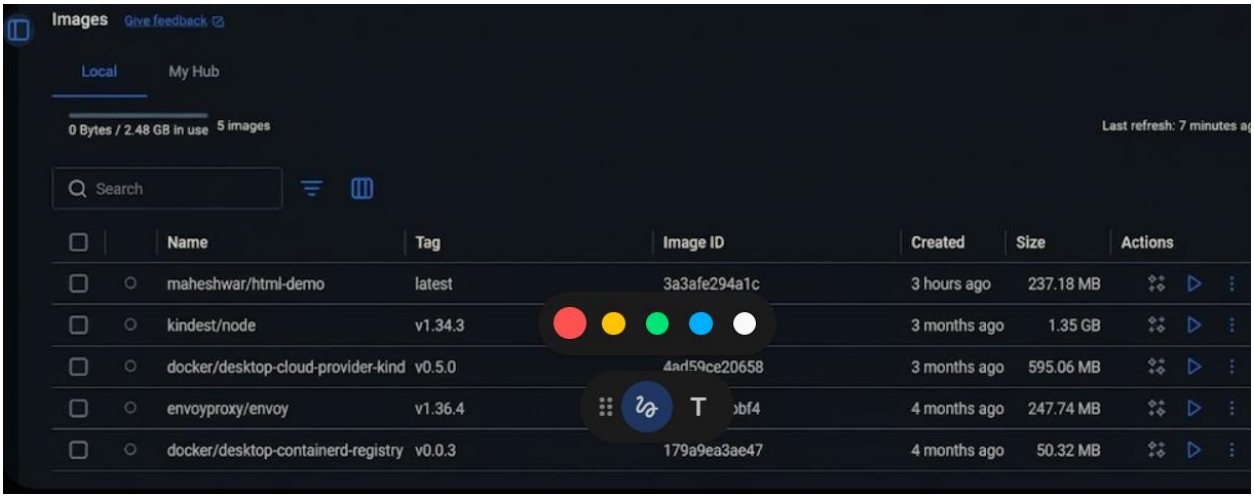
	Clone Code	Build Docker Image	Push Image	Deploy to Kubernetes
Average stage times: (full run time: ~14s)	1s	1s	9s	812ms
#5 Mar 25 18:35 No Changes	1s	1s	9s	812ms

Permalinks

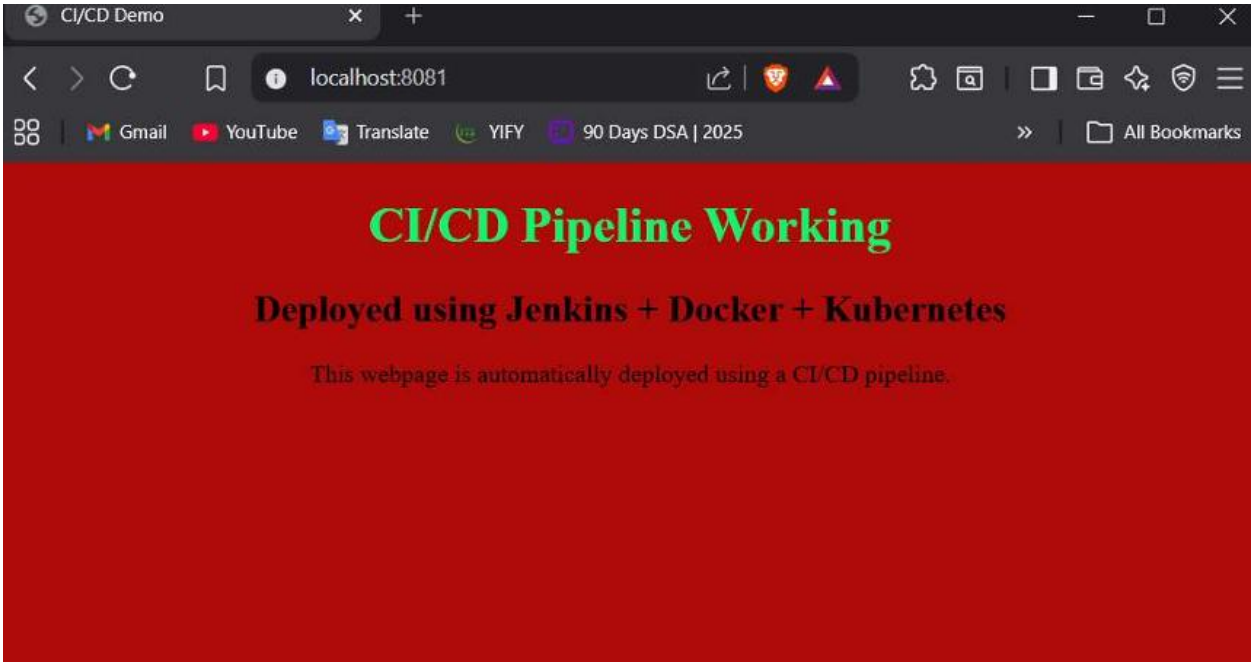
- Last build (#5), 12 min ago
- Last stable build (#5), 12 min ago
- Last successful build (#5), 12 min ago
- Last completed build (#5), 12 min ago



DOCKER-desktop



Node part:8081(after redirecting from 300010)


CI/CD Demo
localhost:8081
Gmail
YouTube
Translate
YIFY
90 Days DSA | 2025
All Bookmarks
CI/CD Pipeline Working
Deployed using Jenkins + Docker + Kubernetes
This webpage is automatically deployed using a CI/CD pipeline.